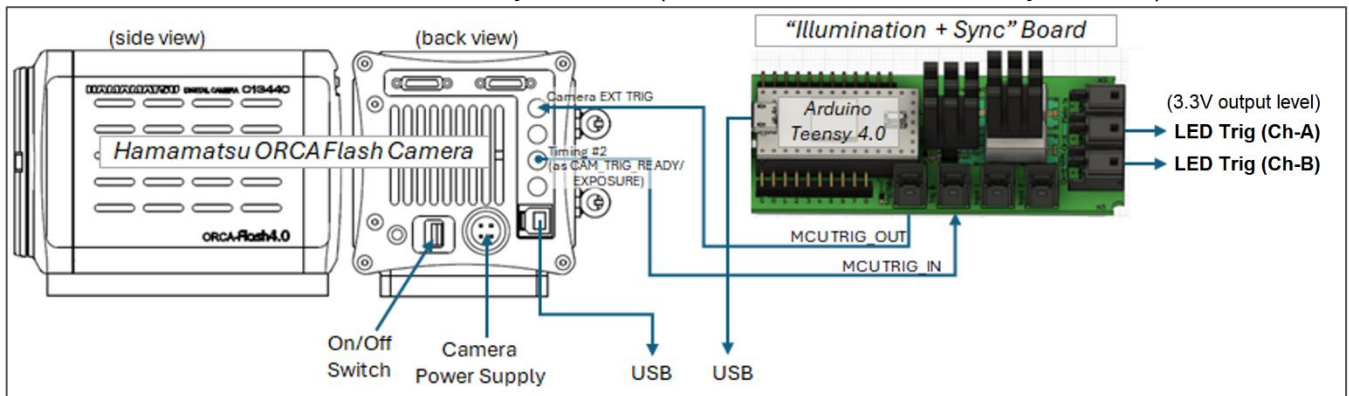


1. Electronic Hardware Connections

The following electronic hardware's are main components of Tilting Microscope.

- Camera : Hamamatsu ORCA-Flash4.0 V3 C13440-20CU
- PCB Module : "Illumination + Sync" Board (with on-board Arduino Teensy 4.0 MCU)



2. Camera

sCMOS camera used for Tilting Microscope

- Camera Model : Hamamatsu ORCA-Flash4.0 V3 C13440-20CU
- Data/Control Interface: USB, 3.3V I/O; EXT TRIG and Timing #2 (as TRIG_READY and EXPOSURE)
- URL : <https://www.hamamatsu.com/jp/en/product/cameras/cmos-cameras/C13440-20CU.html>



Hamamatsu ORCA-Flash4.0 Camera

Main specifications:

- 2048 x 2048 pixels
- 8-bit, 12-bit, 16-bit monochrome image format (12-bit is not supported by Micro-Manager driver)
- 1x1, 2x2 and 4x4 binning
- External Trigger and Internal Trigger
- USB 3.0 interface for control and image data transfer

3. Micro-Manager

Micro-Manager is used to configure camera settings, including exposure, the total number of image frames, image format, and binning. It also manages saving image frames into a stacked TIFF file format.

Additionally, it configures the camera to properly interact with an external "Illumination Driver + Sync" board to run continuous streaming mode and timed illuminations mode for image acquisition.

Download Link: <https://download.micro-manager.org/nightly/2.0/Windows/>

(Time_Illumination.exe control interface was tested with "MMSetup_64bit_2.0.3_20230905.exe")

3.1 Micro-Manager Configuration for Hamamatsu Camera

Steps to create new configure Micro-Manager for Tilting Microscope:

1. Power on Hamamatsu camera and connect it to computer USB interface
2. Launch Micro-Manager
3. When prompt for Hardware Configuration File
 - for the first time, select "(none)"
 - for subsequent launch, just load the desired configuration file and skip all the following steps.
4. Under Micro-Manager Menu, select "Devices | Hardware Configuration Wizard..."
5. Select "Create new configuration", "Next"
6. Under Available Devices, look for HamamatsuHam,
 - Select "HamamatsuHam_DCAM: Universal adapter using Hamamatsu DCAM interface"
 - "Add...". Put the cursor in Device name box and press enter.

(Do wait for a while Micro-Manager to detect the camera.)
7. Select the detected camera (with status "OK"), "Next"
8. "Next", "Next", "Next"
9. "Browse", select folder and enter the name of the configuration file
(eg "C:\Program Files\Micro-Manager-2.0\Timed_Illumination_Hamamatsu_01.cfg")
10. "Finish"

Creation of Grouping (Binning, pixel depth, for Continuous Streaming Mode/Timed Illuminations Mode)

11. Configuration settings, "Group:", "+":
 - Group name, enter "Bit Depth of Pixel"
 - Tick the checkbox "HamamatsuHam_DCM-PixelType"
 - "OK"

12. Configuration settings, “Group:”, “+”:

- Group name, enter “Camera Operation Mode”
- Tick the checkboxes
 - “HamamatsuHam_DCAM-OUTPUT TRIGGER KIND[1]”
 - “HamamatsuHam_DCAM-OUTPUT TRIGGER POLARITY[1]”
 - “HamamatsuHam_DCAM-OUTPUT TRIGGER SOURCE[1]”
 - “HamamatsuHam_DCAM-TRIGGER ACTIVE”
 - “HamamatsuHam_DCAM-TRIGGER SOURCE”
 - “HamamatsuHam_DCAM-Trigger”
 - “HamamatsuHam_DCAM-TriggerPolarity”
- Preset Name : “Continuous Streaming”; (settings are related to I/O interface with external circuit)
- Select the Preset Value as below.

 Preset editor for the "Camera Operation Mode" configuration group

Specify property values for this preset:

Preset name:

Property Name	Preset Value
HamamatsuHam_DCAM-OUTPUT TRIGGER KIND[1]	EXPOSURE
HamamatsuHam_DCAM-OUTPUT TRIGGER POLARITY[1]	POSITIVE
HamamatsuHam_DCAM-OUTPUT TRIGGER SOURCE[1]	TRIGGER
HamamatsuHam_DCAM-TRIGGER ACTIVE	EDGE
HamamatsuHam_DCAM-TRIGGER SOURCE	INTERNAL
HamamatsuHam_DCAM-Trigger	NORMAL
HamamatsuHam_DCAM-TriggerPolarity	POSITIVE

- “OK”

13. Create another preset name:

- Highlight “Camera Operation Mode”
- Preset: “+”
- Preset Name: “Timed Illumination”; (settings are related to I/O interface with external circuit)
- Select the Preset Value as below:

 Preset editor for the "Camera Operation Mode" configuration group

Specify property values for this preset:

Preset name:

Property Name	Preset Value
HamamatsuHam_DCAM-OUTPUT TRIGGER KIND[1]	TRIGGER READY
HamamatsuHam_DCAM-OUTPUT TRIGGER POLARITY[1]	POSITIVE
HamamatsuHam_DCAM-OUTPUT TRIGGER SOURCE[1]	TRIGGER
HamamatsuHam_DCAM-TRIGGER ACTIVE	LEVEL
HamamatsuHam_DCAM-TRIGGER SOURCE	EXTERNAL
HamamatsuHam_DCAM-Trigger	NORMAL
HamamatsuHam_DCAM-TriggerPolarity	POSITIVE

- “OK”

14. “Save” to save the configuration to the same name

(e.g. “C:\Program Files\Micro-Manager-2.0\Timed_Illumination_Hamamatsu_01.cfg”)

Note:

Edit the configuration file directly with text editor to disable the 12-bit pixel depth by deleting the 12-bit setting lines, as the Micro-Manager 2.0.0 driver is not supporting well 12-bit pixel depth.

(Micro-Manager Menu: “Devices”/”Load Hardware Configuration” to load the new configuration file for the changes to take effect.)

```
# Group: Bit Depth of Pixel
# Preset: 8bit
ConfigGroup,Bit Depth of Pixel,8bit,HamamatsuHam_DCAM,PixelFormat,8bit

# Preset: 16bit
ConfigGroup,Bit Depth of Pixel,16bit,HamamatsuHam_DCAM,PixelFormat,16bit

# Preset: 12bit
ConfigGroup,Bit Depth of Pixel,12bit,HamamatsuHam_DCAM,PixelFormat,12bit
```

3.2 Micro-Manager Sequencer Buffer Memory

You may encounter "sequence buffer overflow" (also known as "circular buffer overflow") errors during acquisitions, particularly at high frame rates. To mitigate this, consider increasing the sequence buffer size for fast streaming of large images. Alternatively run the camera at lower pixel depth (8-bit instead of 16-bit) or use binning (2x2 or 4x4 instead of 1x1) to reduce the image size when running fast streaming.

The sequence buffer temporarily stores newly acquired images until the Micro-Manager application is ready to process them (display, save, etc.). It helps manage occasional glitches that slow down software processing, ensuring the camera can maintain its set rate of image acquisition. Note that the sequence buffer is not used for Snap acquisitions or Multi-Dimensional Acquisitions with sufficient time between frames.

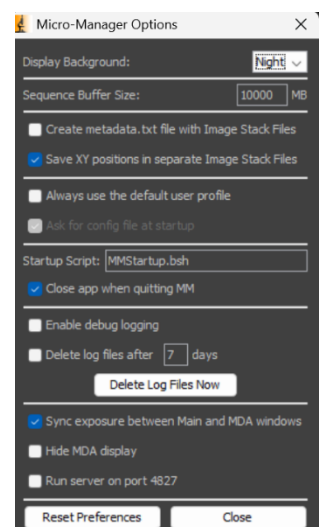
You can adjust the Sequence Buffer Size setting under the Micro-Manager menu (Tools | Options...).

As a general guideline, set the buffer size to 20-100 times the size of a single image from your camera. The buffer should be large enough to accommodate its intended use but small enough to leave RAM available for other tasks and the Operating System.

Be aware that if the sequence buffer size exceeds the available physical RAM + virtual memory, Windows OS will try to increase the virtual memory using more hard disk space. This can significantly increase the launch time of Micro-Manager (potentially one minute or longer). When setting a very large buffer size (e.g., 70000 MB), be patient as Micro-Manager may appear unresponsive during this virtual memory creation. You can monitor memory usage via the Windows Task Manager.

Refer the link below for details.

https://micro-manager.org/Micro-Manager_Configuration_Guide#memory-settings.



3.3 Image Size

The following table gives good estimation of the image size created under different pixel depth and binning settings. One would multiple the image size with the total number of frames to get a good estimate of the file size created. (12-bit pixel depth is not used, as not supported by Micro-Manager 2.0.0 driver.)

Camera pixel size is 2048 x 2048

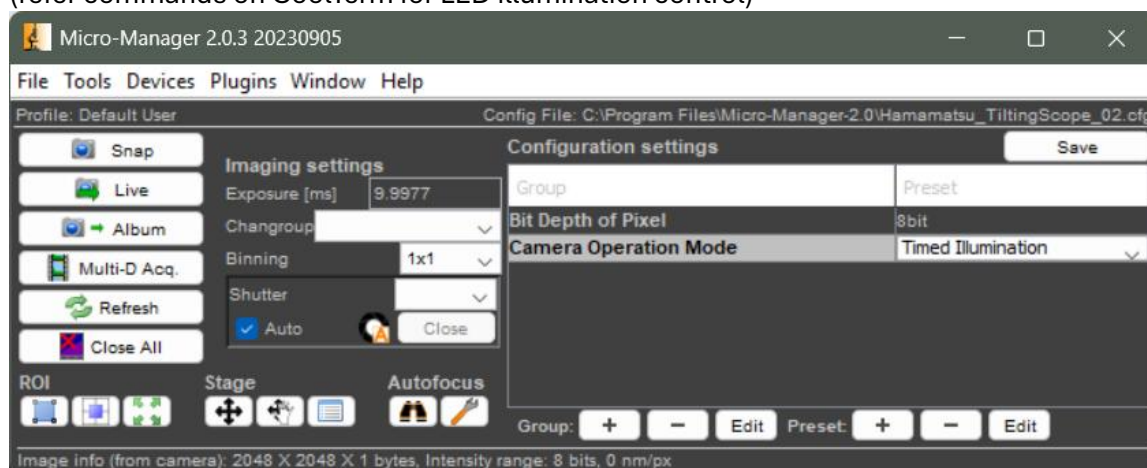
#	Pixel Depth	Binning	Image Size
1	16-bit	1x1	2048 x 2048 x 2-byte (16-bit) = 8.39 MByte
2	16-bit	2x2	1024 x 1024 x 2-byte (16-bit) = 2.10 MByte
3	16-bit	4x4	512 x 512 x 2-byte (16-bit) = 0.524 MByte
4	8-bit	1x1	2048 x 2048 x 1-byte (8-bit) = 4.19 MByte
5	8-bit	2x2	1024 x 1024 x 1-byte (8-bit) = 1.05 MByte
6	8-bit	4x4	512 x 512 x 1-byte (8-bit) = 0.264 MByte

3.4 Live Preview (through Micro-Manager interface)

To preview the camera image:

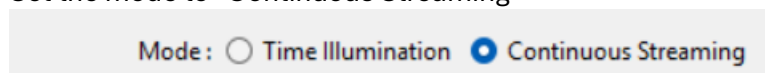
Under Micro-Manager

- configure the “Camera Operating Mode” to “Continuous Streaming”
- Select the desired “Binning” and “Bit Depth of Pixel” settings
- Press “Live” to preview the image.
(refer commands on CoolTerm for LED illumination control)



Under Time_Illumination terminal software

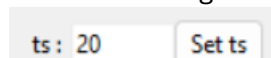
- Connect using the corresponding COM port.
- Set the mode to “Continuous Streaming”



- Set the desired LED illumination channel (one channel at a time)



- Set the ts timing to tally with Micro-Manager Exposure setting, round up to nearest ms.



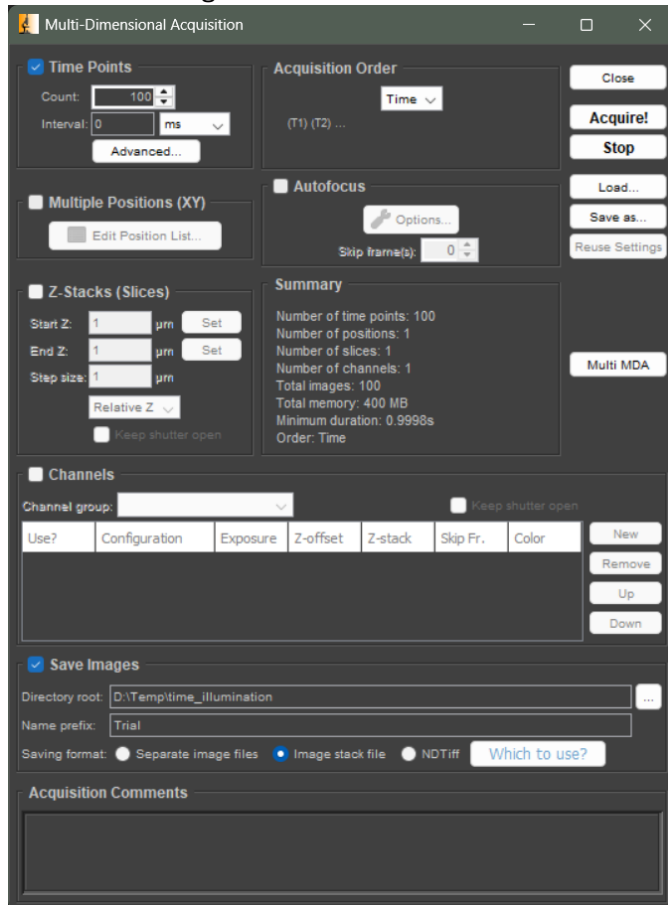
The LED illumination will then automatically turn on and off according to camera exposure, and Live/”Stop Live” operation

3.5 Multi-Dimensional Acquisition with Micro-Manager

Multi-Dimensional Acquisition in Micro-Manager is a powerful feature that allows users to capture complex datasets by varying multiple parameters across different dimensions.

For Tilting Microscope application, configure the following:

- Select “Time Points” checkbox
 - Count : set Total No of Frames to capture
- Save Images (refer below for example illustration)
 - “Directory root”
 - “Name prefix”
 - “Saving format”



To start the acquisition

- Micro-Manager
 - “Count” : 200 (as example)
 - Run command : “Acquire!” button
- Timed_Illumination.exe terminal application
 - Connect to the MCU board using the corresponding COM port
 - Set “Total Cycle” corresponding to half value of Micro-Manager “Count”

Total Cycle: 100 Set Cycles

- Set Mode to “Time Illumination”

Mode: Time Illumination Continuous Streaming

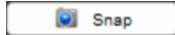
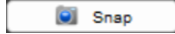
- Press “Start” button

Start
Expected μ -Mgr Point Count: 200

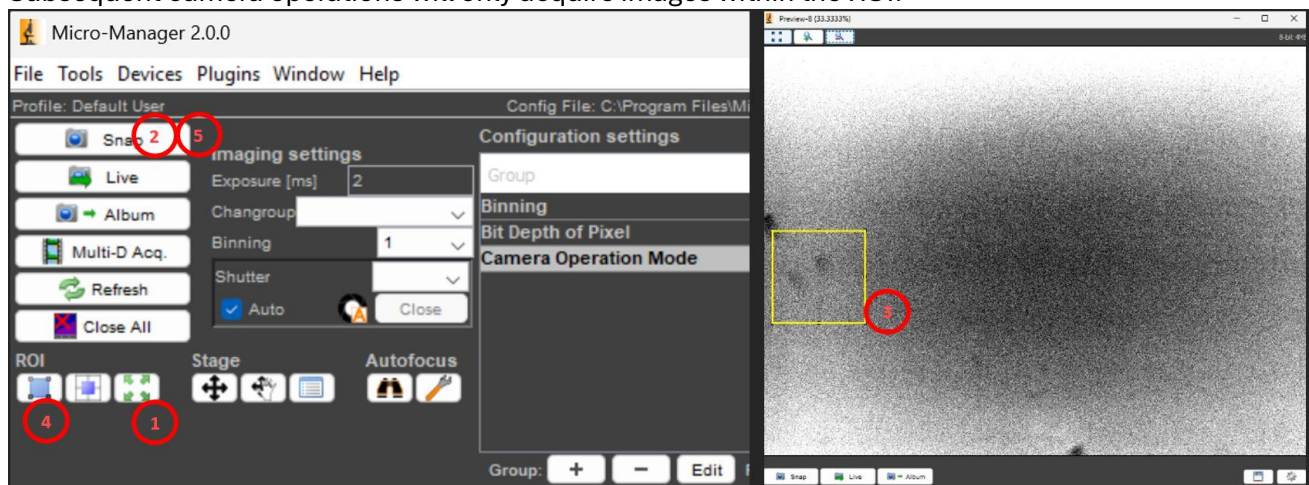
3.6 ROI (Region Of Interest) Operation on Micro-Manager

Follow steps below under Micro-Manager to set the camera to define and acquire images within a region of interest (ROI) .

Enable and Define ROI

1. Set “Camera Operating Mode” to “Continuous Streaming”
2. Press  to select the whole region
3. Press  to take a snap shot of whole region
4. Draw a rectangle over the image to define the desired ROI
5. Press  to enable the ROI mode
6. Press  to take a snap shot of the ROI

Subsequent camera operations will only acquire images within the ROI.



Disable ROI and revert back to full region

1. Press  to select the whole region
2. Press  to take a snap shot of whole region

Subsequent camera operations will acquire images of the whole region.

4. Manual LED illumination operation

For manual LED On/Off operation

- Connect using the corresponding COM port.
- Use these buttons to operate the LED illumination manually

